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PromptForm AI

AI-Powered Form Generation Platform

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Introduction

Online forms run admissions, patient intake, surveys and quizzes, yet every field, type and rule is still built by hand.

PROBLEM
A 10-field form takes **15–25 min**; wrong field types are common and paper or PDF forms must be re-typed.

Objectives

- Prompt → complete form in < 10 s
- Digitise PDF, DOCX, XLSX & images
- 10+ validated field types
- Quizzes, analytics & export
- Secure, team-ready cloud app

Technology Stack

Next.js web framework	**React** UI library
TypeScript language	**Node.js** runtime
Express.js REST API	**PostgreSQL** database
Prisma ORM data access	**Gemini AI** LLM engine

Hosted on Vercel (frontend) and Render (backend).

Proposed Solution

Plain English or a document in; a validated, shareable form out.

1. **Prompt or file** — text, PDF, DOCX, image
2. **AIFormBrain** — plans typed fields
3. **Gemini AI** — schema JSON output
4. **Review & share** — edit, publish, share

Methodology: Agile Scrum · 2-week sprints · 14 weeks · 50 automated tests

Product Demo

1. Plain-English prompt
2. AI-generated live preview
3. Typed, required fields
4. Live analytics

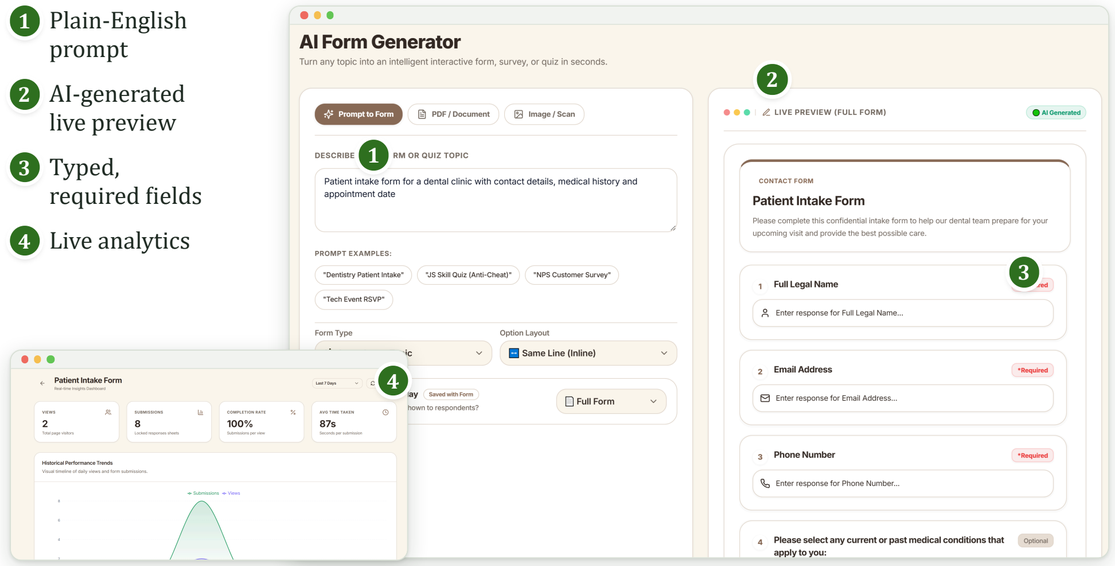


Figure 1. The report's demo prompt generating a six-question Patient Intake Form in the running app; inset: analytics of the demo form (8 submissions, 100 % completion, 87 s average).

System Architecture

User Prompt (text or file) → **Next.js** (frontend) → **Express.js** (REST · JWT) → **Gemini API** (+ AIFormBrain) → **Form JSON** (validated) → **Live Preview** (edit & share)

Data: Express.js → Prisma ORM → PostgreSQL · **Figure 2.** Request flow from prompt to interactive form.

Results & Discussion

50/50	≈ 8 s	10/10
Tests Passed	Form Generation	Objectives Achieved

Performance comparison · time to a ready form

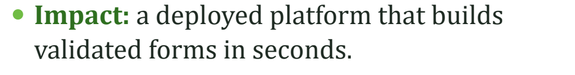


Figure 3. Google Forms, Jotform and PromptForm AI.

Before vs After · building a form

Before · manual builder (15–25 min)
Add fields › Pick types › Set rules › Publish

After · PromptForm AI (≈ 8 s)
Prompt › AI generates › Publish

Figure 4. Field-by-field authoring replaced by one prompt.

- **Reliable:** rule checks fixed wrong types; model fallback kept AI available.
- **Secure:** 9 defects found by API tests, all fixed.

Conclusion

- **Impact:** a deployed platform that builds validated forms in seconds.
- **Applications:** registrations, patient intake, surveys and quizzes.
- **Limitations:** relies on the Gemini API; 30–50 s cold start.
- **Future scope:** payments, AI conditional logic, multilingual input.

References

Acknowledgements

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Scan to Visit PromptForm AI
promptform-ai-frontend.vercel.app